

Body fat composition and distribution in women with polycystic ovary syndrome.

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OBJECTIVES: The aims of the present study were to compare the distribution and accumulation of body fat in women with polycystic ovary syndrome (PCOS) and healthy controls matched for age and body mass index (BMI), and to investigate the association between androgen levels, insulin resistance and fat distribution. **MATERIALS AND METHODS:** Thirty-one PCOS women and 29 age- and BMI-matched healthy control women were evaluated in terms of subcutaneous adipose tissue thickness determined with a skinfold caliper and body composition analyzed by bioelectrical impedance analysis. Blood samples were obtained for determination of follicle-stimulating hormone, luteinizing hormone, 17beta-estradiol, 17-hydroxyprogesterone, basal prolactin, testosterone, dehydroepiandrosterone sulfate, sex hormone-binding globulin (SHBG), androstenedione, insulin and glucose levels. Insulin sensitivity was estimated by fasting glucose/insulin ratio and free androgen index (FAI) was calculated as $100 \times \text{testosterone}/\text{SHBG}$. Differences between means were analyzed by Student's t test or the Mann-Whitney U test according to distribution of the data. Correlation analysis was performed between the body fat distribution and parameters concerning insulin resistance and androgens. **RESULTS:** FAI was significantly higher in patients with PCOS compared with the control group ($p = 0.001$). Fasting insulin was significantly higher and fasting glucose/insulin ratio was significantly lower in the PCOS group vs. controls ($p = 0.03$ and 0.001 , respectively). There was significantly less subcutaneous adipose tissue in the controls than the PCOS women at the triceps ($p = 0.04$) and subscapular region ($p = 0.04$). Waist-to-hip ratio of PCOS women was significantly higher than that of control subjects ($p = 0.04$). **CONCLUSION:** Upper-half type body fat distribution is linked with PCOS, high free testosterone levels and insulin resistance.